Java

Decision making statement, loops

```java
import java.util.Scanner;

public class App {
    public static void main(String[] args) throws Exception {
        System.out.println("Hello, World!");

        //conditional statement
        // if , if else, else if, nested if, switch statement

        int count =10;
        if(count > 10){
            System.out.println("Statement is true");
        }else{
            System.out.println("count is not greater than 10");
        }


        // if(condition){

        // }else if (condition2){

        // }else if(conditon3){

        // }else if(conditon4){

        // }
        int marks = 85;

        if(marks > 90){
            System.out.println("Grade O");
        }else if(marks>80){
            System.out.println("Grade A+");
        }else if(marks>70){
            System.out.println("Grade B");
        }

        // nested if => if inside another if

        int age = 18;
        boolean voterId = true;

        if(age >=18){
            if(voterId){
                System.out.println("you are eligible for voting");
```

```
        }

    }else{
        System.out.println("age is less than 18");
    }


    //switch statement

    int a =1;

    switch(a){
        case 1:
            System.out.println("value is 1");
            break;
        case 2:
            System.out.println("Value is 2");
            break;
        case 3:
            System.out.println("value is 3");
            break;
        default:
            System.out.println("no matching value");
    }

    //loops for, while, do while

    // => we know the iteration count
    for(int i=1; i<=10; i++){
        if(i==5) continue;
        System.out.println(i);
    }


    // while
    // we don't know the iteration count but we know condition

    // boolean isActive = true;
    // int i=1; // intialization
    // while(isActive){
    //     System.out.println(isActive);
    //     if(i== 30){
    //         isActive = false;
    //     }
    //     i++;
    // }
```

```
        //do-while loop

    //    int i = 30;

    //    do{
    //     System.out.println(i);
    //     i--;
    //    }while(i<=5);

    Scanner obj = new Scanner(System.in);

    // int num = obj.nextInt();
    // String str = obj.nextLine();
    // float num2 = obj.nextFloat();
    // boolean isEligible = obj.nextBoolean();
    // char single = obj.next().charAt(0);



    // System.out.println(num);
    // System.out.println(num2);
    // System.out.println(isEligible);
    // System.out.println(single);
    // System.out.println(str);

    //array => contiguous memory location

    int arr[] = {10,20,30};

    System.out.println(arr[0]);
    System.out.println(arr[1]);
    System.out.println(arr[2]);
    // System.out.println(arr[3]);

    }
}
```

Tasks:

Short Answer

1. What is the difference between while and do-while loops?
2. When should you use a switch instead of if-else?
3. What does continue do inside a loop?

Coding Exercises

1. Write a program that checks if a number is **even or odd** using if-else.
2. Use a switch statement to print the day of the week based on a number (1 = Monday, etc.).
3. Create a loop that prints numbers from 1 to 50 but skips multiples of 5.
4. Use a do-while loop to repeatedly ask the user for input until they type "exit".
5. Modify the grading logic to include more ranges (e.g., Grade C, D, F).
6. Create a nested if that checks if a person is eligible for a driving license (age ≥ 18 and has ID).